sequences (Antonov and Saleev 1979) offer a particularly effective way to perform random sampling. Sobol sequences have later been adopted by systems such as SMAC3 (Lindauer et al. 2022).

Random and grid search are both commonly used as baselines, and many popular machine learning libraries include implementations of these simple search techniques [see, e.g., Scikit-learn (Buitinck et al. 2013), Tune (Liaw et al. 2018), Talos (Talos 2019), and H2O (H2O.ai 2017)].

3.2.2 Bayesian optimisation and variants

Evaluating a hyperparameter configuration can be a computationally expensive task since it requires training and validating a machine learning model. Both grid search and random search require a relatively large number of such function evaluations, especially when the number of hyperparameters is large. Bayesian optimisation (Garnett 2023; Hutter et al. 2011; Snoek et al. 2012), sometimes referred to as sequential model-based optimisation, aims to reduce the number of function evaluations by incorporating knowledge about the performance of configurations encountered earlier in the search process. It uses concepts from Bayesian statistics, in particular in the way a statistical model is used to estimate the probability distribution over the possible performance (loss) values of a previously unseen hyperparameter configuration.

Bayesian optimisation utilises knowledge about the search space by fitting a model to the data collected from previously evaluated configurations (the objective function *f*); this model is often referred to as a surrogate model or empirical performance model. The two key ingredients of Bayesian optimisation are the (1) surrogate model and the (2) acquisition function. The surrogate model is a probabilistic model for approximating the objective function *f* that maps hyperparameter configurations to the respective performance values. Bayesian optimisation uses a prior belief of the shape of *f* and updates this prior

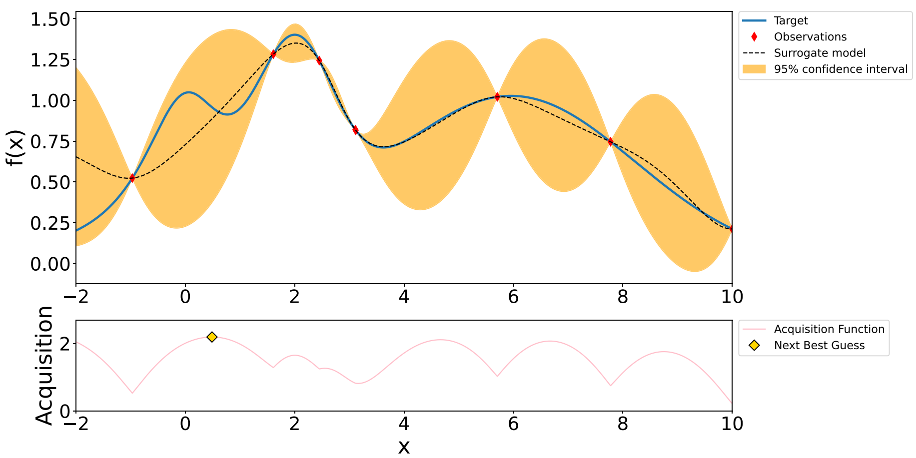

Fig. 4 Bayesian optimisation using a Gaussian process-based surrogate model and the upper confidence bound as acquisition function after seven function evaluations. The solid blue line (target) is the function that is to be optimised, and the dashed line and yellow surface are the mean performance and associated uncertainty derived from the surrogate model, respectively. There is no uncertainty at points where the target function has been evaluated. The acquisition function determines the utility of the configurations to be evaluated (Nogueira 2014). (color figure online)